Track A – Explorer (Beginner to Intermediate)

AI Data Analysis Challenge

Task Document

Objective

The objective of this challenge is to build a simple AI-powered Data Analysis Assistant that can read a CSV dataset, answer predefined questions in natural language, generate one meaningful chart, and explain the findings in a simple and understandable way.

This track is designed for beginners and intermediate students who want to demonstrate their understanding of Python, data analysis, and basic AI integration.

Learning Outcomes

By completing this project, students will learn:

- Reading and processing CSV files using Python.
- Cleaning and understanding datasets.
- Performing basic data analysis.
- Creating visualizations.
- Integrating an LLM (Large Language Model) API.
- Building a simple AI assistant.
- Presenting analytical results professionally.

Project Requirements

Students must create a Python application that performs the following tasks.

Step 1 – Load Dataset

The program should:

- Accept a CSV file.
- Read the dataset successfully.
- Display basic dataset information.

Examples include:

- Number of rows
- Number of columns
- Column names
- Missing values
- Data types

Step 2 – Analyze the Dataset

The application should automatically analyze the dataset using Python.

Students should calculate useful statistics such as:

- Total records
- Average values
- Maximum values
- Minimum values
- Counts
- Frequency
- Category distribution

Students may use:

- Pandas
- NumPy

Step 3 – Answer Natural Language Questions

During the competition, judges will provide **three fixed questions** related to the dataset.

Example questions may include:

- Which product generated the highest sales?
- What is the average age of customers?
- Which city has the maximum orders?
- Which category appears most frequently?

The application should:

- Understand the question.
- Analyze the dataset.
- Return the correct answer.

Students may use:

- Pure Python logic
- One LLM API call
- Rule-based processing

Step 4 – Generate One Chart

The application must generate **at least one meaningful visualization**.

Students can choose any chart depending on the dataset.

Examples:

- Bar Chart
- Pie Chart
- Line Chart
- Histogram
- Scatter Plot

Recommended Libraries:

- Matplotlib
- Plotly
- Seaborn

The chart should include:

- Proper Title
- X-axis Label
- Y-axis Label
- Readable Colors
- Clean Layout

Step 5 – Explain the Result

After generating the chart, the application should provide a short explanation.

Example:

"The Electronics category contributes the highest number of sales, accounting for approximately 42% of the total transactions."

The explanation should be easy to understand.

AI Integration

Students may use:

- OpenAI API
- Gemini API
- Claude API
- Groq API
- DeepSeek API

Only one API call is sufficient.

Agent frameworks are optional.

Project Structure (Suggested)

```
project/
│
├── main.py
├── dataset.csv
├── analysis.py
├── visualization.py
├── requirements.txt
├── README.md
└── charts/
```

User Flow

1. User starts the program.
2. CSV file is loaded.
3. Dataset summary is displayed.
4. Judges ask three questions.

5. Program answers each question.
6. One chart is generated.
7. AI provides a short explanation.

Deliverables

Students must submit:

- Complete Python source code
- requirements.txt
- README.md
- Sample dataset
- Generated chart
- Screenshots (optional)

Judging Criteria

- Correct CSV loading
- Code quality
- Dataset understanding
- Accuracy of answers
- Chart quality
- Explanation quality
- Project organization
- Presentation skills

Bonus Features (Optional)

Students may also include:

- Better UI
- Error handling
- Multiple chart options
- CSV upload button
- Dark mode
- Export chart as PNG
- Export analysis as PDF

These features are optional and will earn bonus appreciation if implemented correctly.

Important Rules

- Python is mandatory.
- CSV dataset must be used.
- Internet should only be used for the selected LLM API.
- The project must run without manual code modification.
- Students should understand every part of their project and be able to explain it during evaluation.

Submission Checklist

☐ Python Project

☐ CSV Dataset

☐ requirements.txt

☐ README.md

☐ Generated Chart

☐ AI Answering System

☐ Project Runs Successfully

☐ Clean Folder Structure

☐ Proper Code Comments